

# Yang Ni

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## 🎓 Education

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| <b>Tsinghua University</b><br>Ph.D. Candidate in Astronomy, Advisor: Prof. Xuening Bai                            | 2022 – Present<br><i>Beijing, China</i>    |
| <b>Princeton University</b><br>Visiting Student Research Collaborators (VSRC), Host Advisor: Prof. Jeremy Goodman | 2026.02 – 2026.08<br><i>New Jersey, US</i> |
| <b>Nanjing University</b><br>B.S. in Astronomy (Magna Cum Laude), Advisor: Prof. Bin Luo                          | 2018 – 2022<br><i>Nanjing, China</i>       |
| <b>Massachusetts Institute of Technology</b><br>Summer Research (remote), Host Advisor: Prof. Mark Vogelsberger   | 2020 – 2021<br><i>Cambridge, US</i>        |

## 🔬 Research Interests

- Early-stage evolution of protoplanetary disks; planet formation
- Star formation in AGN disks
- Evolution of giant molecular clouds; star formation in ISM

## 📄 Publications

(**F** = first author, **C** = co-author)

### First-author papers

- F4.** *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks II. Accretion, Migration, Spin, and Internal Structure of GI Fragments*  
**Y. Ni**, H. Deng, X. Bai      To be submitted, arXiv:2609.37984
- F3.** *Modeling the Evolution of Protoplanetary Disks: Two Pathways from Gravitational Instability to MHD Wind-Driven Accretion*  
**Y. Ni**, W. Xu, X. Bai      re-submitted to ApJ (minor revision), arXiv:2607.12007
- F2.** *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks I. Direct Formation of Gas Giants via Disk Fragmentation*  
**Y. Ni**, H. Deng, X. Bai      ApJ, 995, 96
- F1.** *The life cycle of giant molecular clouds in simulated Milky Way-mass galaxies*  
**Y. Ni**, H. Li, M. Vogelsberger, L. Sales, F. Marinacci, P. Torrey      A&A, 699, A282

### Co-authored papers

- C4.** *MHD-RIGEL: Impact of magnetic field on simulated dwarf galaxies*  
J. Chu, H. Li, Y. Deng, **Y. Ni**      to be submitted
- C3.** *RIGEL: Feedback-regulated cloud-scale star formation efficiency in a simulated dwarf galaxy merger*  
Y. Deng, H. Li, F. Marinacci, **Y. Ni**, B. Liu, A. Smith, R. Kannan, G. L. Bryan      A&A, 704, A240
- C2.** *The THESAN-ZOOM project: Star formation efficiency from giant molecular clouds to galactic scale in high-redshift starbursts*  
Z. Wang, X. Shen, M. Vogelsberger, H. Li, R. Kannan, E. Puchwein, A. Smith, J. Borrow, E. Garaldi, L. Keating, O. Zier, W. McClymont, S. Tacchella, **Y. Ni**, L. Hernquist  
MNRAS, 544, 2675

**C1. GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions**

X. I. Wang, X. Zheng, S. Xiao, J. Yang, Z.-K. Liu, Y.-H. Yang, J.-H. Zou, B.-B. Zhang, M. Zeng, S.-L. Xiong, H. Feng, X.-Y. Song, J. Wen, D. Xu, G.-Y. Chen, **Y. Ni**, Z.-J. Zhang, Y.-X. Wu, C. Cai, J. Cang, Y.-W. Deng, H. Gao, D.-F. Kong, Y. Huang, C.-K. Li, H. Li, X.-B. Li, E.-W. Liang, L. Lin, Y. Liu, X. Long, D. Lu, Q. Luo, Y.-C. Ma, Y.-Z. Meng, W.-X. Peng, R. Qiao, L.-M. Song, Y. Tian, P.-Y. Wang, P. Wang, X.-G. Wang, S. Xu, D. Yang, Y.-H. Yin, W. Zeng, Z. Zeng, T.-J. Zhang, Y. Zhang, Zhao Zhang, Zhen Zhang

ApJ, 922, 237

## Talks and Posters

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**Selected:**

Protoplanetary Disks & Exoplanets 2026, The Kavli Institute for Astronomy and Astrophysics (KIAA) at Peking University, China September 2026

(Invited talk) Purple Mountain Observatory, Jiangsu, China August 2026

AthenaK Summer School, Penn State University, US July 2026

Exoplanets and Planet Formation Mini-Workshop, Tsinghua University, China December 2025

ICEPF Conference 2025, Shanghai, China (*Poster*) December 2025

East Asia Numerical Astrophysics Meeting (EANAM10), Jeju, Korea September 2025

Exoplanets & Planet Formation Workshop 2025, Yunnan, China August 2025

NSFC Scientific Meeting on CPDs and Satellite Formation, Beijing, China May 2024

Exoplanets & Planet Formation Workshop 2023, Beijing, China (*Poster*) December 2023

International Conference on Resolving Galaxy Ecosystems Across All Scales, The Chinese University of Hong Kong, China December 2023

Colloquium, Department of Astrophysics, Universität Zürich, Switzerland November 2023

## Technical Skills

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- ✔ Contributor and experienced user of the multi-grid self-gravity module in AthenaK
- ✔ Contributor and experienced user of the M1-radiative transfer module in GIZMO (PLM reconstruction, RT subcycling, implicit source solver)
- ✔ Experienced user of the radiative (magneto-)hydrodynamic simulation with GIZMO
- ✔ Experienced User of IRAF/Pyraf; The Chandra Interactive Analysis of Observations (CIAO)
- ✔ Experienced User of Python: astropy, h5py, numpy, matplotlib, scipy, yt, networkx... ; C/C++; Mathematica;  $\LaTeX$
- ✔ Experienced User of AI agent harness including Claude Code and Codex
- ✔ Experienced User of LLMs including Claude Opus 4.6/4.7/4.8/5, Claude Fable 5, GPT 5.5/5.6/6, Gemini 3.1 Pro/3.5 Flash

## Honors & Awards

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**Selected:**

Tsinghua University Merit Student, Tsinghua University

Tsinghua Scholarship for Overseas Graduate Studies, Tsinghua University

Comprehensive Scholarship, Tsinghua University

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| National Astronomical Observatory of China (NAOC) Scholarship, | NAOC                                 |
| People's Scholarship First Prize,                              | The Ministry of Education of the PRC |
| Outstanding Undergraduate,                                     | Nanjing University                   |
| "Top-notch" First Prize,                                       | Nanjing University                   |
| G. Zheng Scholarship for Oversea Study,                        | Zheng Foundation                     |

## References

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**(PhD Advisor) Prof. Xuening Bai**, Tsinghua University

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**(PhD Co-Advisor) Prof. Hongping Deng**, Shanghai Astronomical Observatory (SHAO)

✉ [hpdeng353@shao.ac.cn](mailto:hpdeng353@shao.ac.cn)

**Prof. Jeremy Goodman**, Princeton University

✉ [jeremy@astro.princeton.edu](mailto:jeremy@astro.princeton.edu)

**Prof. Mark Vogelsberger**, Massachusetts Institute of Technology

✉ [mvogelsb@mit.edu](mailto:mvogelsb@mit.edu)

**Prof. Hui Li**, Tsinghua University

✉ [hliastro@tsinghua.edu.cn](mailto:hliastro@tsinghua.edu.cn)

**Prof. Bin Luo**, Nanjing University

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